

ibm370\_storage\_sop\_v1json

---

```

{
  "id": "SOP-IBM370-003",
  "title": "IBM System/370 Model 155 Storage Check & Core Isolation",
  "system": "IBM System/370 Model 155",
  "component": "3155 Main Storage / Core Storage Unit",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY33-1057)",
  "description": "Diagnostic procedure for isolating magnetic core storage failures and parity
checks on the Model 155.",
  "steps": [
    {
      "step": 1,
      "action": "Storage Check Indicator Analysis",
      "details": "When a 'Storage Check' occurs, examine the maintenance panel. Check if the
'Storage Select' or 'Storage Logic' error lights are on.",
      "indicator": "Check the CPU Status Register Byte 2, Bit 0 (Storage Parity).",
    },
    {
      "step": 2,
      "action": "Address Identification (MSAR)",
      "details": "Read the Main Storage Address Register (MSAR). This identifies the 8-byte
double-word boundary where the parity failure was detected.",
      "check": "Identify if the error occurred during a 'Fetch' (CPU reading) or 'Store' (CPU
writing) cycle."
    },
    {
      "step": 3,
      "action": "Voltage & Current Marginal Testing",
      "details": "Perform marginal testing on the core drive currents. Reduce the X and Y
drive voltages by 5% using the maintenance switches.",
      "logic": "If errors increase at low voltage, the core array is aging and requires drive
current recalibration or board replacement."
    },
    {
      "step": 4,
      "action": "Core Array Replacement",
      "details": "If the failing address is consistent, isolate to one of the 256K core
storage modules.",
      "remedy": "Physically replace the failing core stack or associated sense-amplifier SLT
card."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Storage Check light solid",
      "next": "Isolate failing address via MSAR."
    },
    {
      "condition": "Intermittent parity during I/O",
      "next": "Check Channel-to-Storage interface timing."
    },
    {
      "condition": "Failures on all bit patterns",
      "next": "Core inhibit driver failure. Replace inhibit board."
    }
  ]
}

```

